

Benhur Tekeste

AI Security · Adversarial Machine Learning · Network and Systems Security

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RESEARCH INTERESTS

AI security and adversarial machine learning; realistic robustness evaluation of ML-based security systems; security of **multi-agent (agentic) AI systems**; AI red teaming; ML for network and systems security.

EDUCATION

Khalifa University

B.Sc. in Computer Engineering – GPA: 3.86/4.00

Abu Dhabi, UAE

Jan 2019 – Dec 2023

PUBLICATIONS

- [C1] **Benhur Tekeste**, et al. “Towards Realistic Adversarial Robustness Evaluation for Intrusion Detection in Industrial Control Systems.” *IEEE International Symposium on Network Computing and Applications (NCA)*, 2026. **Accepted.** (*First author*)
Shows that constraint-blind evaluation overstates the adversarial vulnerability of ML-based ICS intrusion detectors by up to 80–99 percentage points, by isolating in-loop vs. post-hoc constraint enforcement (JSMA, EAD, C&W, PGD; SWaT and WADI testbeds).
- [J1] **Benhur Tekeste**, et al. “Adversarial Machine Learning: A 20-Year Survey of Attacks, Defenses, and Standards.” *IEEE Access*, 2025. (*First author*)
Systematizes two decades of adversarial ML research, covering attack methodologies, defense mechanisms, and the evolving standards landscape for ML security.

RESEARCH EXPERIENCE

Research Assistant – Cybersecurity and AI

Rochester Institute of Technology Dubai

Jul 2025 – Present

Dubai, UAE

- Conduct research on securing AI systems for smart-city applications, with a focus on adversarial machine learning, in collaboration with Dubai Police

Research Assistant – ML-Based Routing for Wireless Ad Hoc Networks

Khalifa University

Feb 2024 – Jan 2025

Abu Dhabi, UAE

- Developed QoS-OLSR, an ML-based routing protocol building on OLSR, to optimize network performance in wireless ad hoc networks
- Built a high-fidelity Mininet-WiFi emulation testbed to validate protocol behavior under realistic wireless constraints
- Integrated QoS-OLSR with Technology Innovation Institute (TII) Comm Sleeve devices running Hardened OS to strengthen the security of the deployed routing stack
- Extended the protocol with IPv6 support, achieving compliance with IPv6 standards
- Mentored undergraduate capstone students implementing a variant of the ML-based QoS-OLSR protocol

ACADEMIC SERVICE

Artifact Evaluation Committee Member, Network and Distributed System Security (NDSS) Symposium

2027

SELECTED RESEARCH PROJECTS

IoT Firmware Reverse Engineering | *Embedded Security, Cryptanalysis, Firmware Analysis*

- Extracted IoT firmware via hardware-assisted ROM dumping and reverse-engineered its embedded cryptographic implementations
- Identified security weaknesses in configuration storage and encryption mechanisms

AWARDS AND RECOGNITION

2 nd Place, Cyber Energy Drill, ADIPEC	2023
3 rd Place, Khalifa University Programming Contest	2023
Sustainability Expert Award, GITEX AI InnovateFest	2023
Distinguished Peer Tutor, Khalifa University	Fall 2023
Inducted, Golden Key International Honour Society	2022

CERTIFICATIONS

Certified Red Team Operator (CRTO), Zero-Point Security	Apr 2025
eLearnSecurity Junior Penetration Tester (eJPT), INE Security	Jun 2024

TECHNICAL SKILLS

Security Domains: Adversarial ML, network security, penetration testing and red teaming, malware and firmware analysis, log forensics

Programming: Python, C, Java, Bash, PowerShell, SQL

ML / AI: TensorFlow, Hugging Face Transformers, LangChain, Pandas, Matplotlib, Gradio

Networking: Mininet-WiFi, OLSR, IPv6, Wireshark

Security Tools: Ghidra, Burp Suite, Metasploit, Cobalt Strike, PowerShell Empire, Nessus, Splunk

Systems: Linux, Windows, Docker, AWS, Git, Flask, L^AT_EX